

This is intended to be part of your "portfolio". More information on portfolios and instructions related to submission can be found at this [link](#).

Recall this design problem from HW4:
 Image you have a macadamia nut that you want to crack open by hand using a simple lever nut cracker. Draw a figure of the nut cracker with the nut when it's about to crack. Calculate the necessary dimensions of the nutcracker and come up with a design to make this task feasible. You can assume a very simple geometry for the nutcracker to make your calculations easier.

Now, instead of relying on grip strength to apply the input force, modify your design to use a linear actuator. Pick a linear actuator from this website, taking into account both force output and size/stroke: <https://www.progressiveautomations.com/collections/linear-actuators>

Given - My HW4 nutcracker design and values
 Handles can now bend
 Find a linear actuator to use.
 location of max deflection
 A1 and material so that δ is less than 2% of its length
 Image of final design
 Upload to Lithub
 Plan: Copy my HW4 design
 Choose a linear actuator

Follow the following steps and questions to update your portfolio page:

Initially, the handles were considered in your design to be rigid. Now, assume the nutcracker handles are no longer rigid. In fact, they are now best described as beams which bend due to the combined action of the forces from the nut and from the actuator.

Consider only the components of these forces transverse to your beam:

- Find the location of maximum elastic deflection in your handles. State your assumptions clearly and describe your analysis.
- Choose a "beam" design (cross-section, material) such that the vertical elastic deflection is below 2% of its length and is the most mass-efficient possible.
- Present your final design in an image or drawing.

Upload your work to your portfolio page, in your GitHub repository.

Please make sure to publish your page and give us a link to the published page instead of a link to the GitHub repository.

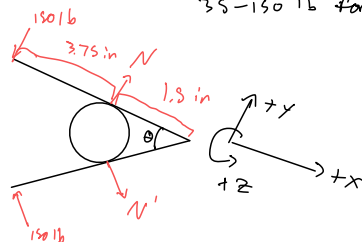
The URL should look like:

cornell-mae-ug.github.io/sp26-portfolio-yourgithubusername/projects/YourProjectName

As a reminder, you can be creative about the layout of the page, but keep your portfolio neat and clean looking. It is publicly accessible and can be viewed by anyone on the internet. As always, be sure to give credit where credit is due.

a- Model as FBD. PA-14 linear actuator

1-14 inch stroke
 35-150 lb force, use 150 lb



AB -

$$\sum M: 150x + M_1 = 0$$

$$M_1(x) = -150x$$

$$\sum M: 150x - 525(x - 3.75) + M_2 = 0$$

$$M_2(x) = 375x - 1968.75$$

$$C_2 = 0 \quad y(0) = 0$$

$$-1054.69 + C_1 = -4746.09 + C_2 \quad y'_1 = y'_2$$

$$-1318.36 + 3.75C_1 + C_2 = -1054.69 + 3.75C_2 + C_4 \quad y_1 = y_2$$

$$0 = -18087.89 + 5.25C_3 + C_4 \quad y(5.25) = 0$$

$$C_3 - C_1 = 3691.4$$

$$C_1 = 632.81$$

$$2460.94 = -C_1 + C_3 + C_4/3.75 \rightarrow \text{matrix} \quad C_3 = 4324.21$$

$$C_4 + 5.25C_3 = 18087.89$$

$$C_4 = -4614.23$$

$$AB - y(x) = \frac{1}{EI} (-25x^3 + 632.81x) \rightarrow GL \rightarrow x = 2.905$$

$$y = 1.18 \times 10^{-6} \text{ m}$$

$$BC - y(x) = \frac{1}{EI} (62.5x^3 - 984.375x^2 + 4324.21x - 4614.23)$$

$$GL \rightarrow x = 3.75$$

$$y = 1.014 \times 10^{-6}$$

Assumed both bars were straight,
 Constant Cross Section,
 Square A1 with $d = 0.5$ in
 Made of steel with $E = 200$ GPa
 Both ends are Pinned

$$\sum F_x: A_x = 0$$

$$\sum F_y: -150 + N + A_y = 0$$

$$N + A_y = 150$$

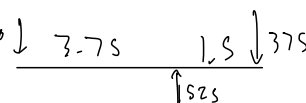
$$\sum M_A: 150(5.25) = N(1.5)$$

$$N = \frac{150 \cdot 5.25}{1.5} = 525 \text{ lb}$$

Plan: rxn's, solve $m(x)$ for segments
 use BC's to find $y(x)$ equations
 Plot both and find max.

$$525 + A_y = 150$$

$$A_y = -375 \text{ N}$$



$$I = \frac{1}{12} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^3 = 0.0052 \text{ in}^4$$

$$AB - \int \int EI y'' = \int -150x$$

$$EI y(x) = \frac{-150x^3}{2 \cdot 3} + C_1x + C_2$$

$$EI y(x) = -25x^3 + C_1x + C_2$$

$$BC's: y_1(3.75) = y_2(3.75)$$

$$y'_1(3.75) = y'_2(3.75)$$

$$y(0) = 0$$

$$EI y(3.75) = -25(3.75)^3 + 3.75C_1 + C_2$$

$$= -1318.36 + 3.75C_1 + C_2$$

$$EI y'(3.75) = -75(3.75)^2 + C_1$$

$$= -1054.69 + C_1$$

$$0 = -25(0)^3 + C_1(0) + C_2$$

$$C_2 = 0$$

$$BC - \int \int EI y'' = \int 375x - 1968.75$$

$$EI y(x) = 62.5x^3 - 984.375x^2 + C_3x + C_4$$

$$BC's: y(5.25) = 0$$

$$y_1(3.75) = y_2(3.75)$$

$$y'_1(3.75) = y'_2(3.75)$$

$$0 = 62.5(5.25)^3 - 984.375(5.25)^2 + 5.25C_3 + C_4$$

$$0 = -18087.89 + 5.25C_3 + C_4$$

$$EI y'(3.75) = 187.5(3.75)^2 - 1968.75(3.75) + C_3$$

$$= -4746.09 + C_3$$

$$EI y(3.75) = 62.5(3.75)^3 - 984.375(3.75)^2 + 3.75C_3 + C_4$$

$$= -1054.69 + 3.75C_3 + C_4$$

x of max deform = 2.905
 and it bends to $y(x) = 1.18 \times 10^{-6}$ in above tx.

$$b - 2\% \text{ of length} = 0.02 \cdot 5.25 = 0.105 = y(x)_{\max}$$

From before, we know $y_{\max}$ occurs at $x = 2.905$ using

the equation $y(x) = \frac{1}{EI} (-25x^3 + 632.81x) \rightarrow$ Find EI such that $y(2.905) = 0.105$

$$0.105 = \frac{1}{EI} (-25(2.905)^3 + 632.81(2.905))$$

$EI = 11620.75 \rightarrow$ Strength for weight, use Ti-6Al-4V $\sim E = 1106 \text{ GPa}$ $\rho = 0.16 \text{ lb/in}^3$

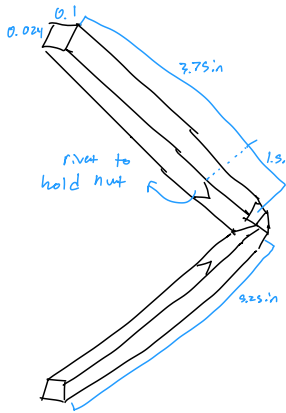
$$I = \frac{11620.75}{110 \times 10^9} = 1.061 \times 10^{-7} \text{ in}^4 = \frac{1}{12} b h^3 \rightarrow \text{want } h > b, \text{ use } b = 0.1 \text{ in}$$

$$1.061 \times 10^{-7} = \frac{1}{12} (0.1) h^3 \rightarrow h = 0.024 \text{ in} \rightarrow 2 \text{ rectangles with } 0.1 \times 0.024 \times 5.25$$

$$V_{\text{total}} = 0.0245 \text{ in}^3$$

$$\frac{0.16 \text{ lb}}{\text{in}^3} (0.0245 \text{ in}^3) = 0.004 \text{ LB}$$

C-



Reflection - Part A made sense, the deflection #'s were good I thought. I think part b is lost a little, 2% of 5.25 inches is a lot of bending and it seems like we may be moving out of the "small deflections"